SAGAR PUBLIC SCHOOL

Integrated Programme • Topic: Work, Power & Energy

Practice Sheet – 1 : ANSWER KEY

Q	Section	Source	Answer	Remark
1	Energy & Conservation	PYQ JEE Main 2025, 3 Apr M	(C) $S/4$, $\sqrt{3gS/2}$	
2	Energy & Conservation	Module S-1 Q28	(i) $x = 2$ m (ii) 16 J	
3	Energy & Conservation	PYQ JEE Main 2026, 6 Apr E	(A) 7.74 m/s	
4	Energy & Conservation	Module O-1 Q12	(B) $mga/2$	
5	Energy & Conservation	PYQ JEE Main 2021, 18 Mar E	6 (m)	
6	Energy & Conservation	Module S-1 Q23	4	uses $g = 10$ m/s^2
7	Energy & Conservation	PYQ JEE Main 2020, 7 Jan M	10 J	
8	Energy & Conservation	Module O-1 Q19	(D) 155.0 J	
9	Work–Energy Thm	PYQ JEE Main 2022, 24 Jun M	(C) 25.0 J	
10	Work–Energy Thm	Module S-1 Q2	(a) 882 J; (b) -247 J; (c) 635 J; (d) 635 J	source key uses $g = 9.8$ m/s^2
11	Work–Energy Thm	PYQ JEE Main 2022, 28 Jun E	(C) 2.0 m/s	
12	Work–Energy Thm	Module O-1 Q8	(D) $(\sqrt{2}+1)$ m/s	
13	Work–Energy Thm	PYQ JEE Main 2026, 2 Apr E	(B) 2020 N	
14	Work–Energy Thm	PYQ JEE Main 2021, 22 Jul E	(C) -627.2 J	
15	Power	PYQ JEE Main 2025, 7 Apr M	(C) $(2t^3 + 3t^5)$ W	
16	Power	Module S-1 Q17	46 J	
17	Power	PYQ JEE Main 2020, 7 Jan E	(D) 66000 W	
18	Vertical Circular	PYQ JEE Main 2024, 29 Jan E	(A) 5 : 1	
19	Vertical Circular	Module S-1 Q32	(a) $\sqrt{2gL}$; (b) $\sqrt{gL}$	
20	Vertical Circular	PYQ JEE Main 2021, 25 Jul M	2 m/s	

Answers are taken from the answer keys printed in the source files (PYQ key / module key). Note: the module key lists a few answers in different notation; numerical answers are in the units stated in the question.